2.4 Parallel & Perpendicular Equations

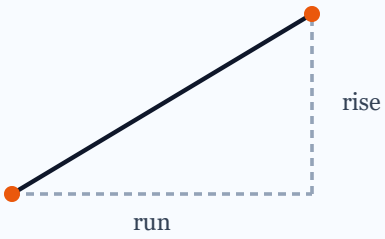
Slope connects the visual idea of parallel and perpendicular lines to hard algebra: equal slopes force lines to run in the same direction forever without meeting, while a slope and its negative reciprocal force a perfect $90°$ turn. This section goes beyond just flipping a fraction — it uses the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ from two raw points, tests whether three points are collinear, and solves for a missing coordinate that makes two lines parallel or perpendicular.

Examples

Example 1: Why parallel lines share slope

A line has slope 3. Explain why any line parallel to it must also have slope 3, then state that slope.

Solution



Slope measures steepness — the rise-to-run ratio never changes along a straight line.

Slope measures a line's steepness and direction; two distinct lines are parallel exactly when they never meet, which requires identical steepness.

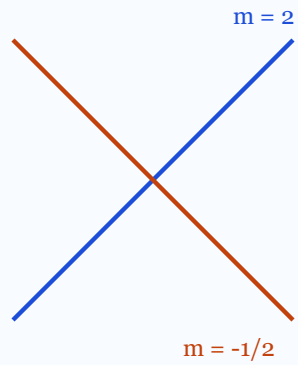
If the slopes differed even slightly, the lines would eventually converge or diverge and cross somewhere.

Answer: The parallel line's slope is also 3.

Example 2: Perpendicular slopes — negative reciprocal

A line has slope 2. Find the slope of a line perpendicular to it, and explain the negative reciprocal rule.

Solution



Perpendicular lines: slopes multiply to −1.

Rule: for two non-vertical, non-horizontal lines, perpendicular slopes multiply to -1: $m_1 \cdot m_2 = -1$.

Solve for the perpendicular slope: $2 \cdot m_2 = -1 \Rightarrow m_2 = -\frac{1}{2}$.

Shortcut: flip the fraction and switch the sign — the "negative reciprocal."

Answer: $-\frac{1}{2}$.

Example 3: Negative reciprocal of a fraction

A line has slope $-\frac{3}{4}$. Find the perpendicular slope.

Solution

Flip the fraction and switch the sign: $-\frac{3}{4} \rightarrow \frac{4}{3}$.

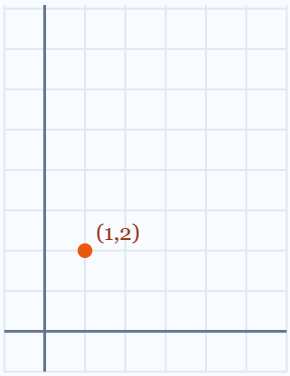
Check: $(-\frac{3}{4})(\frac{4}{3}) = -1$. ✓

Answer: $\frac{4}{3}$.

Example 4: Slope from two points, then a parallel line

Find the slope of the line through $(1, 2)$ and $(5, 10)$, then write the equation of a line parallel to it through $(0, -3)$.

Solution



Two points determine both the slope and the direction of the line.

Apply the slope formula: $m = \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{10 - 2}{5 - 1} = \dfrac{8}{4} = 2.$

A parallel line keeps the same slope, $m = 2$, and uses the new y-intercept -3: $y = 2x - 3$.

Answer: slope $= 2$; parallel line: $y = 2x - 3$.

Example 5: Perpendicular through a point

Write the equation of a line perpendicular to $y = -\dfrac{1}{3}x + 4$ through $(0, 2)$.

Solution

Find the perpendicular slope: negative reciprocal of $-\dfrac{1}{3}$ is 3.

Use the given point as the y-intercept since it has $x = 0$: $b = 2$.

Answer: $y = 3x + 2$.

Example 6: Identify parallel from equations

Are $y = 5x - 1$ and $y = 5x + 7$ parallel?

Solution

Compare slopes directly from slope-intercept form: both have slope 5.

Same slope, different y-intercepts (-1 vs. 7) means the lines are distinct but never meet.

Answer: Yes — parallel.

Example 7: Identify perpendicular from equations

Are $y = 4x$ and $y = -\frac{1}{4}x + 1$ perpendicular?

Solution

Multiply the slopes: $4 \times (-\frac{1}{4}) = -1$.

The product is exactly -1, which is the perpendicular condition.

Answer: Yes — perpendicular.

Example 8: Horizontal and vertical special cases

State the slope of $y = 5$ and the slope of $x = 3$, and explain why the usual product rule can't test them against each other.

Solution

$y = 5$ is horizontal: slope $= 0$ (rise is always 0). $x = 3$ is vertical: slope is undefined (run is always 0, causing division by zero).

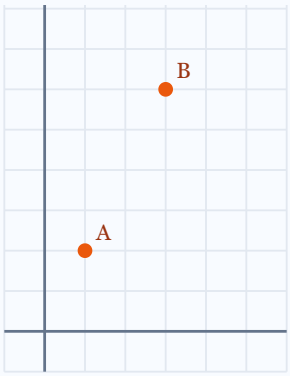
The rule $m_1 m_2 = -1$ needs two actual numbers to multiply — "undefined" isn't a number, so the rule can't be applied directly, even though a horizontal and a vertical line ARE always perpendicular by definition.

Answer: slope of $y = 5$ is 0; slope of $x = 3$ is undefined; they are perpendicular by definition, not by the product rule.

Example 9: Are three points collinear?

Determine whether $A(1, 2)$, $B(3, 6)$, and $C(5, 10)$ are collinear.

Solution



Equal slopes through a shared point force all three onto one line.

Compute the slope between the first two points: $m_{AB} = \dfrac{6-2}{3-1} = \dfrac{4}{2} = 2.$

Compute the slope between the last two points: $m_{BC} = \dfrac{10-6}{5-3} = \dfrac{4}{2} = 2.$

Equal slopes through the shared point B mean all three points lie on the same line.

Answer: Yes, A, B, C are collinear.

Example 10: Solve for a coordinate using perpendicularity

A line through $(2, 1)$ and $(k, 7)$ is perpendicular to a line with slope $-\dfrac{1}{3}$. Find k.

Solution

Perpendicular to $-\dfrac{1}{3}$ means the target slope is the negative reciprocal: 3.

Set the slope formula for the two given points equal to 3: $\dfrac{7-1}{k-2} = 3.$

$$\frac{6}{k-2} = 3 \Rightarrow 6 = 3(k-2) \Rightarrow 6 = 3k - 6 \Rightarrow 12 = 3k \Rightarrow k = 4.$$

Check: slope from $(2, 1)$ to $(4, 7)$ is $\dfrac{6}{2} = 3$, and $3 \times (-\dfrac{1}{3}) = -1$. ✓

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Parallel slope

Slope of line parallel to slope 7?

Answer: **Verification:**

Problem 2: Perp slope

Perpendicular to slope 5?

Answer: **Verification:**

Problem 3: Perp slope

Perpendicular to slope $-\frac{2}{3}$?

Answer: **Verification:**

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Problem 4: Parallel equation

Parallel to $y = -x + 4$ through $(0, 1)$.

Answer: **Verification:**

Problem 5: Perp equation

Perpendicular to $y = \frac{1}{2}x$ through $(0, -3)$.

Answer: **Verification:**

Problem 6: Parallel?

$y = 3x + 2$ and $y = 3x - 5$ parallel?

Answer: **Verification:**

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Problem 7: Perpendicular?

$y = 2x$ and $y = 2x + 1$ perpendicular?

Answer: **Verification:**

Problem 8: Perpendicular?

$y = -\frac{1}{4}x$ and $y = 4x + 3$ perpendicular?

Answer: **Verification:**

Problem 9: Horizontal slope

Slope of $y = 5$?

Answer: **Verification:**

Problem 10: Vertical

Equation of vertical line through $x = 3$?

Answer: **Verification:**

Problem 11: Perp to horizontal

Line perpendicular to $y = 2$?

Answer: **Verification:**

Problem 12: Slope product

If lines are perpendicular (neither vertical), product of slopes?

Answer: **Verification:**

Problem 13: Write parallel

Parallel to $y = 4x - 1$ through $(0, 0)$.

Answer: **Verification:**

Problem 14: Challenge — collinearity

Determine whether $A(-2, -1)$, $B(1, 5)$, and $C(3, 9)$ are collinear, showing the slope work.

Answer: **Verification:**

Problem 15: Challenge — solve for a coordinate

A line through $(1, 4)$ and $(m, 10)$ is parallel to $y = \frac{3}{2}x - 1$. Find m.

Answer: **Verification:**

✓ Answer Key

1. 7
2. $-\frac{1}{5}$
3. $\frac{3}{2}$
4. $y = -x + 1$
5. $y = -2x - 3$
6. Yes
7. No
8. Yes
9. 0
10. $x = 3$
11. A vertical line
12. -1
13. $y = 4x$
14. Yes; slope $AB = 2 =$ slope BC
15. $m = 5$

Great work — keep practising!